

(h) Chemical tests	
Students should:	
2.44	describe tests for these gases: - hydrogen - oxygen - carbon dioxide - ammonia - chlorine.
2.45	describe how to carry out a flame test
2.46	know the colours formed in flame tests for these cations: Li^+ is red Na^+ is yellow K^+ is lilac Ca^{2+} is orange-red Cu^{2+} is blue-green.
2.47	describe tests for these cations: NH_4^+ using sodium hydroxide solution and identifying the gas evolved Cu^{2+}, Fe^{2+} and Fe^{3+} using sodium hydroxide solution.
2.48	describe tests for these anions: Cl^-, Br^- and I^- using acidified silver nitrate solution SO_4^{2-} using acidified barium chloride solution CO_3^{2-} using hydrochloric acid and identifying the gas evolved.
2.49	describe a test for the presence of water using anhydrous copper(II) sulfate
2.50	describe a physical test to show whether a sample of water is pure

3 Physical chemistry

The following sub-topics are covered in this section:

- (a) Energetics
- (b) Rates of reaction
- (c) Reversible reactions and equilibria